

Problem 1:

Let R be a ring. Let $I_1, \dots, I_n \subset R$ be ideals. Let $\pi : R \rightarrow \prod_{i=1}^n R/I_i$ be the homomorphism given by $x \rightarrow (\pi_1(x), \dots, \pi_n(x))$, where $\pi_i : R \rightarrow R/I_i$ is the quotient map. We will say that the ideals I_i are pairwise coprime if for all $i \neq j$ the ideals I_i and I_j are coprime.

(a)

Prove that if the ideals I_i are pairwise coprime, then $I_1 I_2 \dots I_n = I_1 \cap I_2 \cap \dots \cap I_n$

Proof.

Base Case:

Assume I, J are coprime.

This implies there exists $x \in I$ and $y \in J$ s.t. $x + y = 1$. Thus, let $a \in I \cap J \implies ax + ay = a \implies a \in IJ$. Thus, $I \cap J = IJ$.

Inductive Case:

Assume this works for $k \leq n$. Consider $n + 1$.

Also since I_n and I_{n+1} are coprime this implies that $I_n I_{n+1} = I_n \cap I_{n+1}$.

Let $j \neq n, n + 1$. Consider $I_j + I_n I_{n+1}$. Since $I_j + I_n = 1$ and $I_j + I_{n+1} = 1$ this implies that there exists $a, b \in I_j$ and $x \in I_n$ and $y \in I_{n+1}$ that $a + x = 1$ and $b + y = 1$. Thus, $(a + x)(b + y) = 1 \implies ab + ay + bx + xy = 1$. Since $ab + ay + bx \in I_j$ and $xy \in I_n I_{n+1} \implies I_j + I_n I_{n+1} = 1$.

Thus, by induction hypothesis $I_1 \cap \dots \cap I_n I_{n+1} = I_1 I_2 \dots I_{n+1}$. ■

(b)

Give an example to show that the equality $I_1 I_2 \dots I_n = I_1 \cap I_2 \cap \dots \cap I_n$ can fail if the ideals I_i are not pairwise coprime.

Proof.

Consider $\mathbb{Z}$. Consider the ideals (12) and (8) . First $(12) \cap (8) = (4)$, and $(12)(8) = (24)$. ■

(c)

Prove that $\pi : R \rightarrow \prod_{i=1}^n R/I_i = S$ is surjective if and only if I_i are pairwise coprime.

Proof.

Assume π is surjective. Let $e^{(i)} \in S$ s.t. it has a 1 in the i th spot and 0 everywhere else. Since surjective let $e_i \in R$ s.t. $\pi(e_i) = e^{(i)}$.

Since $e_i \equiv 1 \pmod{I_i} \implies 1 - e_i \in I_i$.

Also, for $i \neq j$ $e_i \in I_j$ since $\pi_j(e_i) = 0$.

This implies that $1 = (1 - e_i) + e_i \in I_i + I_j$. Therefore, I_i 's are pairwise coprime.

Assume I_i are pairwise coprime.

For $i \neq j, k$ and $j \neq k$ this implies that $I_i + I_j = 1$ and $I_i + I_k = 1$. Thus, there exists $a, b \in I_i, x \in I_j$, and $y \in I_k$ s.t. $a + x = 1$ and $b + y = 1$. Thus, $ab + ay + bx + xy = 1$. This implies that $I_i + I_j I_k = 1$. Thus, inducting on all $j \neq i$, you can show that $I_i + \prod_{i \neq j} I_j = 1$. Thus, for each i there exists $a_i \in I_i$ and $b_i \in \prod_{i \neq j} I_j$ s.t. $a_i + b_i = 1$. This implies that $b_i \equiv 1 \pmod{I_i}$ and $b_i \in I_j$ for $i \neq j$.

Thus, given $(\bar{x}_1, \dots, \bar{x}_n) \in S$ consider $x = \sum_{j=1}^n b_j x_j \in R$.

Then for each k we get that $x \equiv b_k x_k \equiv x_k \pmod{I_k}$ since $b_k \equiv 1 \pmod{I_k}$ and $b_k \in I_k$ for $k \neq i$.

Thus, $\pi(x) = (\bar{x}_1, \dots, \bar{x}_n)$. ■

(d)

Prove that π is injective if and only if $I_1 \cap \dots \cap I_n = 0$.

Proof.

Assume π is injective.

Let $x \in I_1 \cap \dots \cap I_n$. This implies that $\pi_i(x) = 0$ since $x \in I_i$ thus $\pi(x) = 0 \in \ker(\pi)$ thus since π is injective this implies $x = 0$. Thus, $I_1 \cap \dots \cap I_n = 0$.

Assume $I_1 \cap \dots \cap I_n = 0$.

Let $x \in \ker(\pi)$. This implies that $\pi(x) = 0$. This implies that $\pi_i(x) = 0 \implies x \in I_i$ for all i . Thus, $x \in I_1 \cap \dots \cap I_n$ this implies $x = 0$. Thus, $\ker(\pi) = 0 \implies \pi$ is injective. ■

Problem 2:

Let R be a noetherian ring. For an R -module M , prove that the following conditions are equivalent:

- M has finite length.
- M is a finitely generated R -module and $\dim \operatorname{Supp}_R(M) \leq 0$.

Proof.

Assume M has finite length.

If $M = 0$ then $\dim \operatorname{Supp}_R(M) = -\infty$.

If $M \neq 0$

Shown in class this implies M is noetherian and artinian thus, M is finitely generated R -module.

Since M has finite length it has a composition series

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$$

Shown in class $M_i/M_{i-1} \cong R/m_i$ for some maximal ideal m_i .

Also shown in class since $0 \rightarrow M_{i-1} \rightarrow M_i \rightarrow M_i/M_{i-1} \rightarrow 0$ is a short exact sequence we get that

$$\operatorname{Supp}(M_i) = \operatorname{Supp}(M_{i-1}) \cup \operatorname{Supp}(M_i/M_{i-1}) = \operatorname{Supp}(M_{i-1}) \cup \operatorname{Supp}(R/m_i)$$

Thus, $\operatorname{Supp}(M) = \bigcup_{i=1}^n \operatorname{Supp}(M_i/M_{i-1})$, but since $(R/m)_{\mathfrak{p}} \cong R_{\mathfrak{p}}/mR_{\mathfrak{p}} \neq 0 \iff m \subset \mathfrak{p}$ this implies that $m = \mathfrak{p}$ thus, $\operatorname{Supp}(R/m) = \{m\}$.

Therefore, $\operatorname{Supp}(M)$ is finite set of maximal ideals, so the dimension is 0.

Assume M is finitely generated and $\dim \operatorname{Supp}_R(M) \leq 0$.

If $M = 0$ then M is finite length.

If $M \neq 0$

Since M is finitely generated this implies that $\operatorname{Supp}(M) = V(\operatorname{Ann}(M))$. Therefore, $\dim \operatorname{Supp}(M) = \dim \operatorname{Spec}(R/\operatorname{Ann}(M)) = \dim(R/\operatorname{Ann}(M)) \leq 0$.

Since R is noetherian this implies that $R/\operatorname{Ann}(M)$ is noetherian, and therefore, it is artinian since it is dimension 0.

Since we can view M as a finitely generated $R/\operatorname{Ann}(M)$ module and we know that this quotient is artinian this implies that M is artinian and noetherian. Thus, M has finite length. ■

Problem 3:

Let R an artinian ring. For an R -module M , prove that the following conditions are equivalent:

- (a) M is finitely generated R -module.
- (b) M is an artinian R -module.
- (c) M is a noetherian R -module.
- (d) M is finite length R -module.

Proof.

(a) $\implies$ (b), (c) $\implies$ (d) by proofs in class.

(c) $\implies$ (a) by definition of noetherian.

(d) $\implies$ (b), (c) $\implies$ (a) by proofs in class.

(b) $\implies$ (a)

Assume not (a). Thus, M is not finitely generated.

Let N be a countably infinite submodule. This implies that $N = (n_1, n_2, \dots)$.

Consider $(n_1, n_2, \dots) \supsetneq (n_2, n_3, \dots) \supsetneq \dots$

Thus, N is not artinian since this does not stabilize. Thus, M is not artinian since N is not.

Thus, (b) $\implies$ (a). ■

Problem 4:

For a ring R , prove that the following conditions are equivalent:

- (a) R is artinian.
- (b) R is noetherian and $\text{Spec}(R)$ is discrete and finite.
- (c) R is noetherian and $\text{Spec}(R)$ is discrete.

Also, give an example of a ring R such that $\text{Spec}(R)$ is discrete and finite, but R is not artinian.

Proof.

(a) $\implies$ (b) in class we have shown that R being artinian implies R is noetherian and $\text{Spec}(R) = \text{mSpec}(R)$ is finite.

Consider $\mathfrak{m} \in \text{Spec}(R)$ since $\mathfrak{m}$ is maximal this implies that $V(\mathfrak{m}) = \{\mathfrak{m}\}$. Thus, all singletons are closed which implies every set is closed since we only have a finite amount of elements. Thus, every set is open. Thus, $\text{Spec}(R)$ is discrete.

(b) $\implies$ (c) trivial.

(c) $\implies$ (a)

Assume (c) since $\text{Spec}(R)$ is discrete this implies that every set is closed. Thus, all singletons are closed. This implies for all $\mathfrak{p} \in \text{Spec}(R)$ that $V(\mathfrak{p}) = \{\mathfrak{p}\}$. This implies that no other prime ideal contains $\mathfrak{p}$. This, implies that $\mathfrak{p}$ is maximal. Thus, $\text{Spec}(R)$ only has maximal ideals, so $\dim \text{Spec}(R) = 0$. Thus, since R is noetherian and dimension zero this implies R is artinian.

Consider $R = \mathbb{C}[x_1, x_2, \dots] / (x_1^2, x_2^2, \dots)$

$\text{Spec}(R) = \{(2), (3)\}$

Consider $\sqrt{(0)}$ since $x_i^2 = 0 \implies x_i \in \sqrt{(0)}$ thus, $(x_1, x_2, \dots) \subset \sqrt{(0)}$.

If $f \in (x_1, x_2, \dots)$ then f contains no constant term. Thus, $f^2 = 0$, so $f \in \sqrt{(0)}$. Therefore, the nilradical is a maximal ideal. Since the nilradical is in the intersection of all prime ideals this implies that the nilradical is the only prime ideal since it is maximal.

Thus, $\text{Spec}(R)$ is discrete and finite, but it is not artinian since

$(x_1, x_2, \dots) \supsetneq (x_2, x_3, \dots) \supsetneq (x_3, x_4, \dots) \supsetneq \dots$ ■

Problem 5:

Give an example of a ring R such that $\dim R = 0$ and $\operatorname{Spec}(R)$ is infinite.

Proof.

Consider $R = \prod_{i=1}^{\infty} \mathbb{F}_2$.

Shown in homework 1 this has all maximal ideals and has an infinite amount. Thus,
 $\dim R = 0$ ■

Problem 6:

Let R be an artinian ring. Prove that there is an isomorphism

$$R \cong \prod_{\mathfrak{m} \in \text{Spec}(R)} R_{\mathfrak{m}}$$

Proof.

Shown in class since R is artinian this implies there exists an integer $k > 0$ s.t. $R \cong \prod_{\mathfrak{m} \in \text{Spec}(R)} R/\mathfrak{m}^k$

Consider $\pi : R \rightarrow R_{\mathfrak{m}}$

Since $\mathfrak{m}^k = \mathfrak{m}^{k+1} \implies \mathfrak{m}^k R_{\mathfrak{m}} = \mathfrak{m}^{k+1} R_{\mathfrak{m}} \implies \mathfrak{m}^k R_{\mathfrak{m}} = 0$ by Nakayama's lemma.

Thus, for $x \in \mathfrak{m}^k \implies \pi(x) = 0 \implies x \in \ker(\pi) \implies \mathfrak{m}^k \subset \ker(\pi)$.

Let $x \in \ker(\pi) \implies sx = 0$ for $s \notin \mathfrak{m}$.

Assume for contradiction that $(s) + \mathfrak{m}^k \neq R$ this implies $(s) + \mathfrak{m}^k \subset n$ for some maximal ideal n .

However, since $\mathfrak{m}^k \subset (s) + \mathfrak{m}^k \subset n \implies \mathfrak{m} \subset n \implies \mathfrak{m} = n$.

Thus, $s + a^k = n$ for $a, n \in \mathfrak{m}$. This implies that $s = n - a^k \in \mathfrak{m}$ this is a contradiction.

Thus, $(s) + \mathfrak{m}^k = R$. This implies there exists $a \in R$ and $b \in \mathfrak{m}^k$ s.t. $as + b = 1 \implies xas + xb = x \implies x = xb \in \mathfrak{m}^k$.

Therefore, $\ker(\pi) = \mathfrak{m}^k$.

Let $r/s \in R_{\mathfrak{m}}$. As above pick $a \in R$ and $b \in \mathfrak{m}^k$ s.t. $as + b = 1$.

In $R_{\mathfrak{m}}$ we have $b/1 = 0$ thus, $(a/1)(s/1) = 1$, so $s^{-1} = a/1$.

Thus, $r/s = (r/1)s^{-1} = (r/1)(a/1) = (ra/1) = \pi(ra)$. Thus, π is surjective.

Therefore, by the first isomorphism theorem we get that $R_{\mathfrak{m}} \cong R/\mathfrak{m}^k$.

Thus, from the above isomorphism we get that $R \cong \prod_{\mathfrak{m} \in \text{Spec}(R)} R_{\mathfrak{m}}$. ■

Problem 7:

Let R be an artinian ring.

(a)

Let $(0) = q_1 \cap \cdots \cap q_n$ be a minimal primary decomposition. Prove that there is an isomorphism

$$R \cong \prod_{i=1}^n R/q_i$$

Proof.

Since R is artinian this implies that $\sqrt{q_i} = m_i$ for some maximal ideal m_i .

Consider $q_i \subset m$ for some $m \in \text{Spec}(R)$. This implies $m_i \subset m$ thus, $m_i = m$. This implies that the only prime ideal that contains q_i is m_i .

Thus, R/q_i is an artinian local ring. This implies that jacobson radical is nilpotent.

Thus, $J = m_i/q_i$ and there exists a $k_i > 0$ s.t. $(m_i/q_i)^{k_i} = 0 \implies m_i^{k_i} \subset q_i$.

Thus, for $i \neq j$ since $m_i + m_j = R$ we get that $m_i^{k_i} + m_j^{k_j} = R$ by a similar proof in the previous problem.

This, implies $R = m_i^{k_i} + m_j^{k_j} \subset q_i + q_j$. Therefore, all q_i are pairwise coprime.

Also since the intersection is zero by problem one we get that $R \cong \prod_{i=1}^n R/q_i$. ■

(b)

Suppose that $R \cong \prod_{i=1}^n R_i$ and $R \cong \prod_{j=1}^m R'_j$ are isomorphisms, where R_i and R'_j are artinian local rings. Prove that $n = m$ and there exists a permutation σ of $\{1, \dots, n\}$ s.t. $R_i = R_{\sigma(i)}'$ for all $1 \leq i \leq n$.

Proof.

Let $q_i = \ker(R \rightarrow R_i)$. Since $\ker(R \rightarrow \prod_i R_i) = \cap \ker(R \rightarrow R_i) = \{0\}$ this implies that $\cap q_i = (0)$.

Since $\varphi : R \rightarrow R_i$ is surjective this implies that $r \in \sqrt{\ker(\varphi)} \iff \varphi(r)^n = \varphi(r^n) = 0 \iff \varphi(r) \in \sqrt{(0)}$.

Thus, since R is artinian and local this implies that $\sqrt{(0)} = m_{R_i}$ the unique prime ideal in R_i .

Thus, $m_i = \sqrt{q_i} = \varphi^{-1}(m_{R_i})$.

This implies that q_i is m_i -primary since its radical is a maximal ideal m_i .

Consider $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0) \in \prod_{i=1}^n R_i$ and $e_j \in \prod_{i=1}^m R_i$ where there is a 1 in

the i th spot for e_i and 0 everywhere else. Same for e_j .

Since there is an isomorphism this implies there exists $a, b \in R$ s.t. $\phi(a) = e_i$ and $\phi(b) = e_j$.

This implies that $a \notin \sqrt{q_i} = m_i$ since e_i has a 1 in the i th coord and $1 \notin m_i$, and $a \in \sqrt{q_{i'}} = m_{i'}$ for $i \neq i'$ since 0 is in the i' th coord.

Thus, $\sqrt{q_i} \neq \sqrt{q_{i'}}$. Therefore, all $\sqrt{q_i}$ are distinct. Also, for q_i consider $\cap_{i \neq k} q_k$. This is not a subset set of q_i since if it was this would implies that $\sqrt{\cap_{i \neq k} q_k} \subset \sqrt{q_i} \implies \cap_{i \neq k} m_k \subset m_i$. This implies that $m_j = m_i$ for some $j \neq i$, but this is a contradiction since $\sqrt{q_i} \neq \sqrt{q_j}$ for $i \neq j$.

Thus, this is also not redundant intersection.

Since all the primes in $\{\sqrt{q_i} = m_i\}$ are maximal this implies they are all also minimal in that set. Thus, shown in class we know that all the m_i are uniquely determined by (0).

Thus, by a similar proof $q'_i = \ker(R \rightarrow R'_i)$ are also a minimal primary decomposition for (0) this implies that $q_i = q'_{\sigma(i)}$. Thus, $m = n$.

Also, from the first isomorphism theroem since each $R \rightarrow R_i$ is surjective we get that $R/q_i \cong R_i$. Therefore, since $q_i = q'_{\sigma(i)}$ this implies that $R_i \cong R'_{\sigma(i)}$. ■